FABIAN RADENTA BANGUN

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EDUCATION

Institut Teknologi Bandung — *Bandung, Indonesia*
Bachelor of Engineering in Informatics, Artificial Intelligence specialization — *Aug 2022 – Aug 2026*

RESEARCH AND ENGINEERING

Selective Tiling with a Gate Classifier for Small Object Detection on Edge Devices — *Bandung, Indonesia*
Undergraduate final project, grade A | PyTorch, ONNX Runtime, YOLO11n, MobileNetV2, Raspberry Pi 4 — *Oct 2025 – Aug 2026*

- Cut per-image inference time by 51.6% on a Raspberry Pi 4, from 20.8 seconds down to 9.6 seconds, by routing 4K drone imagery through a MobileNetV2 gate classifier that passes only the 877 of 2,009 tiles predicted to contain objects on to a YOLO11n detector, while holding small-object AP flat at 0.310 against 0.308 for full tiling.
- Grounded the design in on-device profiling, measuring the gate at 72 ms per tile against the detector's 1,522 ms per tile, so screening costs only 4.7% of what it saves.
- Trained the MobileNetV2 gate to 91.9% validation accuracy on binary tile classification, and tuned the decision threshold to trade missed tiles against tiles skipped.
- Benchmarked YOLOv8n, YOLO11n, and YOLOv10n under identical conditions and selected YOLO11n on AP-small (0.157 against 0.136 and 0.146) with 14% fewer parameters than YOLOv8n, then trained it on 8,017 tiles to mAP50 0.839.
- Deployed the pipeline on-device with ONNX Runtime after testing INT8 weight-only quantization across 26,533 detections and rejecting it, since detector confidence collapsed to a 0.502 ceiling even though the model shrank from 10.28 MB to 2.94 MB.
- Ran the full ablation across 512, 1024, and 2048 pixel tile sizes on the UAVVaste dataset of 772 4K drone images, showing that no single tile size wins on every metric.

PROFESSIONAL EXPERIENCE

Mioto, Software Engineer Intern (Mobile) — *Jakarta, Indonesia*
IoT company specializing in connected mobility | sole mobile developer on a 3-person team — Jul – Sep 2025

- Built the company's Flutter field-data collection app for Android and iOS, taking it from an empty repository to a client-facing demo within a 10-week internship.
- Engineered bidirectional Bluetooth Low Energy communication with GPS tracker hardware, covering device scanning, MTU negotiation, characteristic notifications, and a custom command and response protocol, so the app could read telemetry and geofence data directly from devices in the field.
- Designed an offline-first data layer in SQLite that buffers telemetry locally and syncs to the backend in batches of 400, keeping the app fully usable with no connectivity. In internal testing a full device upload finished in under 5 minutes, against the multi-day manual workflow the company reported for the same task.
- Integrated 11 REST endpoints with JWT authentication, secure token storage, request timeouts, and automatic logout on unauthorized responses.
- Shipped QR-based device-to-worker pairing, CSV export, and route playback with distance, duration, and average speed calculation.
- Configured native Android and iOS permissions and networking requirements, and built signed release APKs for on-device field testing.

AWARDS

1st Place in the VTOL Category and Best Strategy Award | Certificate — *Lampung, Indonesia*
Kontes Robot Terbang Indonesia 2023 — *Mar – Sep 2023*

- One of 18 national finalists, each from a different university. Adapted and extended the team's OpenCV vision modules within the Robotic Software Control division to meet requirements for autonomous UAV target detection.

PROJECTS

Flagrail, Feature Flag Management Platform (solo) | Github — *May 2026 – Present*
Personal project | Next.js, TypeScript, PostgreSQL, Prisma, Vitest | github.com/fabianradenta/flagrail

- Built a multi-tenant feature-flag platform of roughly 5,000 lines of TypeScript on a 9-model PostgreSQL schema covering projects, members, per-environment flag state, targeting rules, hashed API keys, and an audit log.
- Designed the evaluation engine as a pure function with no I/O that returns one of 8 explicit reason codes, and gave it deterministic bucketing over the SHA-256 of the flag key and user id, so a given user sees the same variant on every request and across deploys.
- Secured a public evaluation API with bearer keys stored only as SHA-256 hashes and surfaced once on creation, and hand-rolled JWT sessions using jose in HTTP-only cookies without an authentication framework.
- Covered the engine with 18 Vitest unit tests, including bucketing regression guards and a statistical check that a 50% rollout reaches roughly half of a large user sample.
- Shipped a live consumer demo that calls the evaluation API and swaps its checkout interface in real time, exposing the full evaluation breakdown of audience match, computed bucket, rollout outcome, and reason.

Travel Trip, Website Product for Open Trip Operators (2-person project) | Github | Demo — *Aug 2026 – Present*

Astro, TypeScript, Cloudflare Workers, GitHub Actions | travel-trip.radentafabian.workers.dev

- Built and deployed a live static site product for open-trip operators on Cloudflare Workers, driven by a GitHub Actions pipeline with push, daily cron, and manual triggers, pinned to fixed Node and Wrangler versions so builds stay reproducible over time.
- Solved a staleness problem inherent to the static build, where departures are filtered by date at build time, by scheduling a daily rebuild that keeps expired trips off the live site without introducing a backend.
- Packaged the site as two commercial tiers behind a single feature flag, where the premium build exposes departure dates, seat availability, and month filters, and the basic build hides everything that would otherwise need an admin panel to maintain.
- Modelled trip content as typed Astro collections validated with Zod, attaching price and remaining seats to each departure rather than to the package, since high-season batches are priced differently.
- Owned the scheduling and filtering features, the content data layer, and the build and deploy setup across a two-person team.

Information Retrieval System with GAN Query Expansion | [Github](https://github.com/Ariel-HS/Tubes_STBI) *Jun 2026*

Course project | Python, Streamlit | https://github.com/Ariel-HS/Tubes_STBI

- Built an interactive document retrieval application using TF-IDF and cosine similarity, combined with a GAN-based query expansion component.
- Implemented caching and state management so machine learning models and document indices are not reloaded on every query, and added a two-stage query validation workflow.

LinkinPurry, Job Portal Platform | [Github](https://github.com/Labpro-21/if3110-tubes-2024-k02-21/) *Oct – Nov 2024*

Course project | PHP (MVC), PostgreSQL, Docker | https://github.com/Labpro-21/if3110-tubes-2024-k02-21/

- Built server-side CRUD modules for User and Job Application using native PHP in an MVC architecture, with PostgreSQL for persistence.
- Containerized the application with Docker Compose for a reproducible local setup, and exposed the endpoints consumed by the frontend.
- Collaborated through Git branch, pull request, and merge workflows in a shared team repository.

Dobot Sorting Automation System *Oct 2025 – Jan 2026*

Industrial IoT course project | Revolution Pi

- Engineered the remote HMI and local network communication layer for a robotic sorting cell, configuring real-time target-zone execution on industrial edge computing hardware.

SEKA, Interactive Educational Mannequin *May 2025*

Commissioned project | Embedded C/C++

- Built and delivered a paid prototype of an interactive mannequin that teaches body-safety awareness to students with special needs, implementing the instructional design of a special education researcher at Universitas Pendidikan Indonesia.
- Programmed an ESP32 in C/C++ to read 10 touch sensors and trigger matching voice clips through a DFPlayer Mini with microSD storage, so the mannequin tells the student which parts of the body are appropriate for others to touch and which are not.

AjarinDong, Mobile Learning App Concept | [Deck](https://drive.google.com/file/d/1jKxQ6OSoGbpeWT3SfRR4u5qf3W0x-WPs/view) *Mar – Sep 2023*

UI/UX course project | Figma | https://drive.google.com/file/d/1jKxQ6OSoGbpeWT3SfRR4u5qf3W0x-WPs/view

- Designed user flows, wireframes, and high-fidelity mobile interfaces in Figma, and ran user research and usability testing to refine the design.

LEADERSHIP

HMIF ITB, Informatics Student Association *Bandung, Indonesia*

Head of Community Service *May 2025 – Mar 2026*

- Led a 17-person division across three program lines covering village community development, volunteer teaching, and charity distribution, reaching more than 50 direct beneficiaries.
- Introduced introductory computational thinking classes for local students as part of the division's educational outreach.

SKILLS

ML & CV	PyTorch, YOLO (v8, v10, v11), MobileNet, OpenCV, ONNX Runtime, object detection, edge deployment, model benchmarking
Mobile	Flutter, Dart, Bluetooth Low Energy, SQLite, offline-first sync, Android and iOS release builds
Web	TypeScript, JavaScript, React, Next.js, Prisma, Tailwind CSS, REST API, PHP
Languages	Python, TypeScript, JavaScript, Dart, C/C++, SQL
Data	PostgreSQL, MySQL, SQLite
Embedded & IoT	Raspberry Pi 4, ESP32, Arduino, Revolution Pi
Tools	Git, Docker, Figma, Streamlit